

Entropy-Regularized Optimal Transport and Schrödinger Bridges

Optimal Transport and Stochastic Analysis · Lecture 2 of 3

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Recall: Lecture 1

Recall: couplings and the Monge–Kantorovich problem

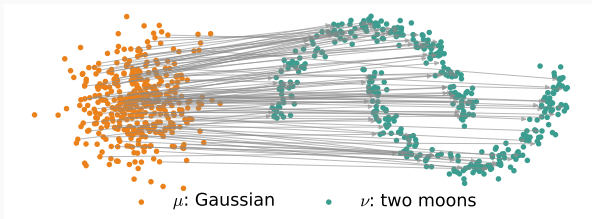
- ▶ A **coupling** of (μ, ν) : a pair of random vectors (X, Y) with $X \sim \mu$, $Y \sim \nu$. Its law $\gamma \in \Pi(\mu, \nu)$.
- ▶ OT with quadratic cost:

$$\mathbb{W}_2^2(\mu, \nu) = \inf_{\gamma \in \Pi(\mu, \nu)} \mathbb{E}_\gamma \|X - Y\|^2.$$

- ▶ $\mathbb{W}_2$ is a metric on probabilities with finite second moments.
- ▶ If $X \sim \mu$ has a density, the optimal coupling is **unique** and is a Monge solution

$$Y = \nabla \varphi(X), \quad \varphi \text{ convex.}$$

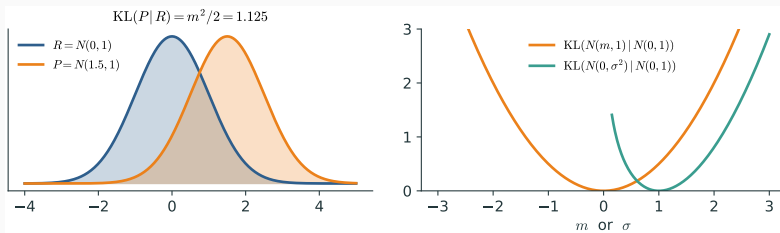
Recall: optimal couplings are degenerate



- ▶ Y is a **deterministic** function of X . No randomness is left in the coupling.
- ▶ Computing it exactly is expensive; estimating it from samples is statistically hard.
- ▶ Today: **add a little randomness** to the coupling on purpose.
- ▶ We will give two algorithms then for efficient computation.

Relative entropy (Kullback–Leibler)

- ▶ $P \ll R$: $\text{KL}(P | R) := \mathbb{E}_P \left[\log \frac{dP}{dR}(X) \right] \geq 0$, with equality iff $P = R$ (Jensen).
 $+\infty$ if $P \not\ll R$.
- ▶ Example: $\text{KL}(\text{U}(2, 3) | \text{U}(0, 1)) = +\infty$: disjoint supports.



Why the sudden interest in OT in statistics, ML, AI?

- ▶ More robust than Kullback–Leibler: $\mathbb{W}_2(\mu, \nu) < \infty$ even for disjoint supports,

$$\text{KL}(\text{U}(2, 3) \mid \text{U}(0, 1)) = \infty, \quad \mathbb{W}_2(\text{U}(2, 3), \text{U}(0, 1)) = 2.$$

- ▶ Any problem with an underlying geometry: $\mathbb{W}_2(\delta_x, \delta_y) = \|y - x\|$.
- ▶ Uncoupled regression, single-cell genomics, matching, vision, sampling, image generation.
- ▶ **But** expensive to compute, and statistically cursed by dimension.
- ▶ Solution is to regularize with entropy.

See Chewi, Niles-Weed & Rigollet, *Statistical Optimal Transport* (lecture notes).

Static Schrödinger bridge

Entropy

- ▶ $X \sim \mu$ with density f :

$$\text{Ent}(\mu) := \mathbb{E}[\log f(X)].$$

$\text{Ent}(\mu) = +\infty$ if μ has no density.

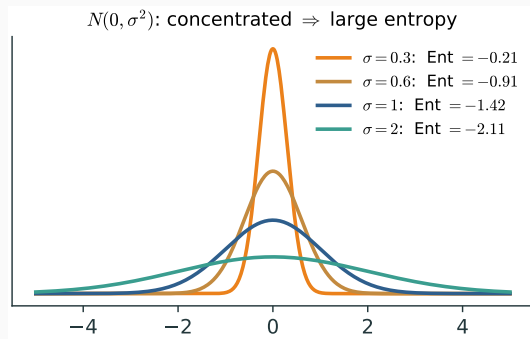
- ▶ Example: $X \sim N(0, \sigma^2)$,

$$\text{Ent} = -\log \sigma - \frac{1}{2} \log(2\pi e).$$

- ▶ Concentrated laws have **large** entropy.

Moral

Optimal couplings $(X, T(X))$ has **infinite** entropy.



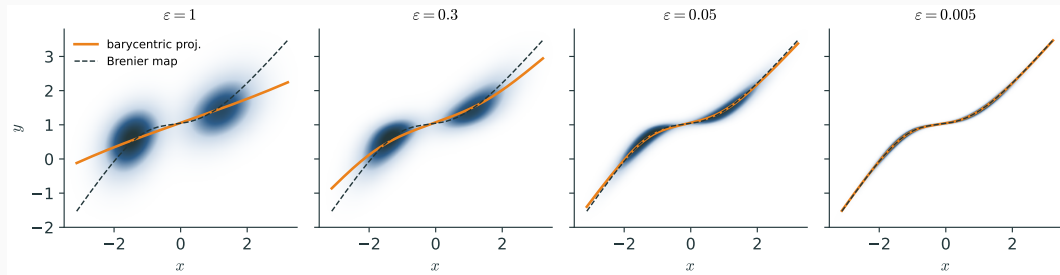
- ▶ Galichon & Salanié '09, Cuturi '13: Regularize OT with entropy,

$$\text{EOT}_\varepsilon(\mu, \nu) := \inf_{\gamma \in \Pi(\mu, \nu)} \left\{ \frac{1}{2} \mathbb{E}_\gamma [\|X - Y\|^2] + \varepsilon \text{Ent}(\gamma) \right\}.$$

- ▶ Transport cost pulls towards Brenier, while entropy pulls away.
- ▶ unique minimizer γ_ε : the (static) Schrödinger bridge at temperature ε .
- ▶ As $\varepsilon \rightarrow 0$: $\gamma_\varepsilon \rightarrow$ Brenier coupling.

Galichon & Salanié (2009); Cuturi (2013).

Trading geometry for probability



- The same μ (bimodal) and ν (Gaussian) as Lecture 1. Shade: the density of γ_ε .
- Orange: $x \mapsto \mathbb{E}_{\gamma_\varepsilon}[Y | X = x]$, Entropic Brenier map. Dashed: Brenier map. They merge as $\varepsilon \downarrow 0$.

Theorem (Csiszár '75, Rüschendorf & Thomsen '93)

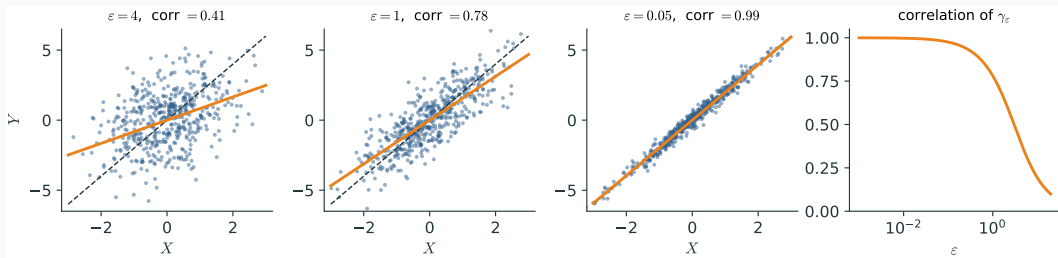
If $\text{EOT}_\varepsilon(\mu, \nu) < \infty$, there are $\phi_\varepsilon, \psi_\varepsilon : \mathbb{R}^d \rightarrow \mathbb{R}$ with

$$\gamma_\varepsilon(x, y) = \exp \left(\frac{1}{\varepsilon} \left[\phi_\varepsilon(x) + \psi_\varepsilon(y) - \frac{1}{2} \|x - y\|^2 \right] \right) \mu(x) \nu(y).$$

The **Schrödinger potentials** $(\phi_\varepsilon, \psi_\varepsilon)$ are unique up to $(\phi_\varepsilon + a, \psi_\varepsilon - a)$.

- The potentials are pinned down by the marginal constraints and are a solution of a fixed point equation.

Gaussians in pictures



- ▶ $d = 1$, $X \sim N(0, 1)$, $Y \sim N(0, 4)$. γ_ϵ is jointly Gaussian with $\text{Cov}(X, Y) = c_\epsilon = \frac{\sqrt{16 + \epsilon^2} - \epsilon}{2}$.
- ▶ Samples of $(X, Y) \sim \gamma_\epsilon$. Dashed: Brenier map $y = 2x$.
- ▶ Orange: Entropic Brenier map $\mathbb{E}_{\gamma_\epsilon}[Y | X = x] = c_\epsilon x$, which tends to $2x$.

Multivariate closed forms: Janati et al. (2020); Mallasto et al. (2022).

The entropic Brenier map is a gradient of a convex function

1. Fact: the function below is always convex.

$$\mathbb{E}_{\gamma_\varepsilon}[Y | X = x] = x - \nabla \phi_\varepsilon(x),$$

and the Hessian is $\frac{1}{\varepsilon} \text{Cov}_{\gamma_\varepsilon}(Y | X = x) \succeq 0$.

Entropic Brenier map

Like Brenier's map, the entropic map is a gradient of a convex function; it converges to the Brenier map as $\varepsilon \rightarrow 0$.

Pooladian & Niles-Weed (2021). See the my talk tomorrow for the error.

Theorem (Mikami '04, Léonard '12)

As $\varepsilon \rightarrow 0^+$:

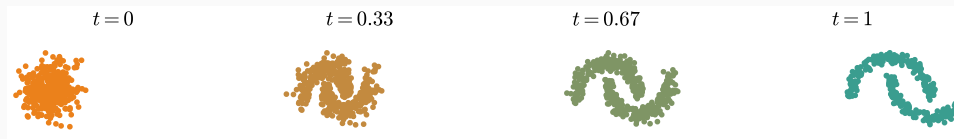
$$\text{EOT}_\varepsilon(\mu, \nu) \rightarrow \frac{1}{2} \mathbb{W}_2^2(\mu, \nu), \quad \gamma_\varepsilon \rightarrow \text{Brenier coupling.}$$

Recall: McCann interpolation

- ▶ $X_0 \sim \mu$, $X_1 = \nabla\varphi(X_0) \sim \nu$. Walk in straight lines:

$$X_t = (1 - t)X_0 + tX_1, \quad \rho_t = \text{Law}(X_t).$$

- ▶ Curve $t \mapsto \rho_t$, $t \in [0, 1]$, is McCann interpolation joining μ to ν .



- ▶ **Theorem** (P. '19; Conforti & Tamanini '21), under regularity assumptions:

$$\text{EOT}_\varepsilon(\mu, \nu) = \frac{1}{2}\mathbb{W}_2^2 + \frac{\varepsilon}{2}(\text{Ent}\mu + \text{Ent}\nu) + \frac{\varepsilon^2}{8} \int_0^1 I(\rho_t) dt + o(\varepsilon^2).$$

- ▶ $I(\rho_t) = \mathbb{E}_{\rho_t} \|\nabla \log \rho_t\|^2$ is the Fisher Information of ρ_t .
- ▶ The leading error depends only on the marginal. The next one does not.
- ▶ (P. '19) Conditional density of Schrödinger bridge is approximately Gaussian.

$$Y \mid X = x \approx N(\nabla \varphi(x), \varepsilon \nabla^2 \varphi(x)).$$

The correction to Brenier map

- **Theorem**(Agarwal-Harchaoui-Mulcahy-P. '24, Mulcahy-P. '26) Under assumptions on μ, ν

$$\mathbb{E}_{\gamma_\varepsilon}[Y \mid X = x] = \nabla\varphi(x) + \frac{\varepsilon}{2}\nabla\log\mu(x) + o(\varepsilon), \quad \text{in } L^2(\mu).$$

$$\mathbb{E}_{\gamma_\varepsilon}[X \mid Y = y] = \nabla\varphi^*(y) + \frac{\varepsilon}{2}\nabla\log\nu(y) + o(\varepsilon), \quad \text{in } L^2(\nu).$$

- Again, the leading error depends only on the marginal.

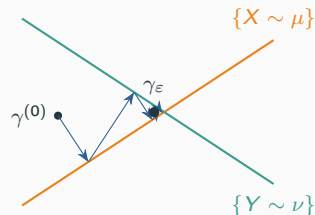
How to compute Schrödinger
bridge? Algorithm 1: Sinkhorn

Sinkhorn / IPFP: alternating projections

- ▶ Start from $\gamma^{(0)} \propto e^{-\|y-x\|^2/2\varepsilon} \mu \otimes \nu$. Alternately fix one marginal by a KL-projection (Csiszár '75).
- ▶ Each step gives a joint density

$$\gamma^{(k)} = e^{-\|y-x\|^2/2\varepsilon - \phi_k(x)/\varepsilon - \psi_k(y)/\varepsilon} \mu \otimes \nu.$$

- ▶ Fortet ('40) used it to solve the Schrödinger system. Also Deming & Stephan ('40), Sinkhorn ('64).



Iterates keep the product form; the marginals are only right in the limit.

Sinkhorn by hand

- ▶ $\epsilon = 1$. μ, ν uniform on $\mathcal{X} = \{0, 1\}$, $\mathcal{Y} = \{0, -1\}$. Initialize:

$$\begin{bmatrix} 1 & e^{-1/2} \\ e^{-1/2} & e^{-2} \end{bmatrix}.$$

- ▶ Make row sums $(1/2, 1/2)$.

$$\begin{bmatrix} \frac{1}{2}(1 + e^{-1/2})^{-1} & \frac{1}{2}e^{-1/2}(1 + e^{-1/2})^{-1} \\ \frac{1}{2}(e^{-1/2} + e^{-2})^{-1}e^{-1/2} & \frac{1}{2}(e^{-1/2} + e^{-2})^{-1}e^{-2} \end{bmatrix} \approx \begin{bmatrix} 0.3 & 0.2 \\ 0.4 & 0.1 \end{bmatrix}.$$

- ▶ Make column sums $(1/2, 1/2)$.

$$\begin{bmatrix} 3/14 & 1/3 \\ 4/14 & 1/6 \end{bmatrix}$$

- ▶ And so on

The Sinkhorn revolution

- ▶ Solving OT on n data points is a linear program: complexity $\tilde{O}(n^3)$.
- ▶ Galichon & Salanié '09, Cuturi '13 proposed the **Sinkhorn algorithm** for EOT_ϵ .
- ▶ Each iteration is two matrix–vector products: highly parallelisable on GPUs.
- ▶ (Altschuler, Weed & Rigollet '17) Near-linear time $\tilde{O}(n^2)$ for a fixed accuracy.
- ▶ Differentiable in the data

Exponential convergence for $\varepsilon > 0$

- ▶ Finite (or compactly supported) problems: exponential convergence for fixed $\varepsilon > 0$, by contraction in Hilbert's projective metric (Birkhoff '57; Franklin & Lorenz '89).
- ▶ Unbounded supports with tail conditions: Conforti, Durmus & Greco '23; Ghosal & Nutz '22; Eckstein '23.
- ▶ Typical rates: $C_\varepsilon \kappa_\varepsilon^n$, $C_\varepsilon > 0$, $\kappa_\varepsilon \in (0, 1)$, $n = \text{iteration}$.
- ▶ Under assumptions, as $\varepsilon \downarrow 0$, uniform rates $C\kappa^{n/\varepsilon}$. Chiarini et al. '24, Chizat et al. '24.
- ▶ Low temperature: Deb, Kim, P. & Schiebinger '23 show that Sinkhorn becomes a **Wasserstein mirror gradient flow** in the limit. Also see Berman '20.

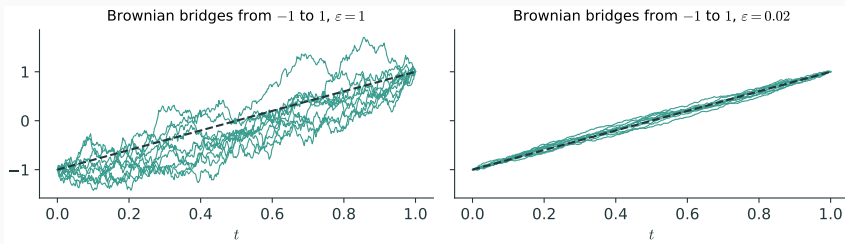
Dynamic regularisation on path space

Brownian bridges

- ▶ R^ε - Brownian motion in $\mathbb{R}^d$, variance ε .
- ▶ Brownian bridge, $R^{\varepsilon,xy}$: Brownian motion (variance ε) from x conditioned to be at y at time 1. With BB a standard Brownian bridge from 0 to 0,

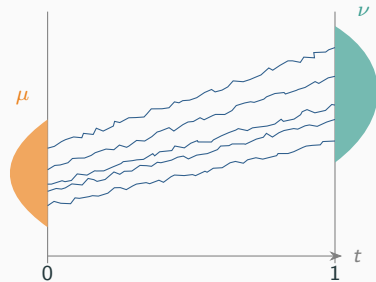
$$X_t = (1-t)x + ty + \sqrt{\varepsilon} BB_t, \quad \mathbb{E}X_t = (1-t)x + ty, \quad \text{Var}(X_t) = \varepsilon t(1-t).$$

- ▶ The mean path is the straight line. The noise shrinks as $\varepsilon \rightarrow 0$.



Schrödinger's lazy gas experiment (1931)

- ▶ N independent Brownian gas molecules, variance ε , in a **very cold** room.
- ▶ Their empirical law is $\approx \mu$ at $t = 0$ and, unexpectedly, $\approx \nu$ at $t = 1$.
- ▶ What's their most likely path?
- ▶ Answer: Dynamic Schrödinger bridge (Föllmer '88).



Schrödinger (1931, 1932); Föllmer (1988); Léonard (2014) survey.

Dynamic Schrödinger bridge

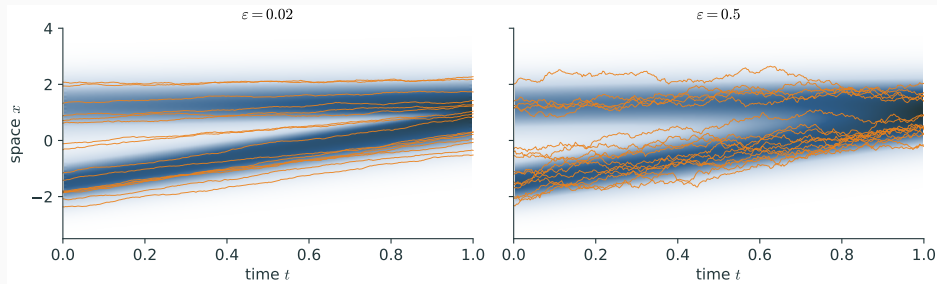
1. Sample (X, Y) from the static Schrödinger bridge. Join them by ε Brownian bridge. I.e.,

$P^\star = \text{"draw } (X_0, X_1) \sim \gamma_\varepsilon, \text{ then join them by a Brownian bridge"}.$

2. The molecules “know” what the Brenier map is ahead of time.

The dynamic Schrödinger bridge is a **mixture of Brownian bridges**, mixed by the static one. Example of a so-called **reciprocal process**. It is also a **Markov process**!

Pictures of Schrödinger bridges



- Shade: density of X_t under P^* . Orange: sample paths, drawn exactly by the recipe on the previous slide.
- Small ε : close to the McCann interpolation. Large ε : blurred, paths cross freely.

How to compute Schrödinger
bridge? Algorithm 2: **Iterated
Markov Projections**

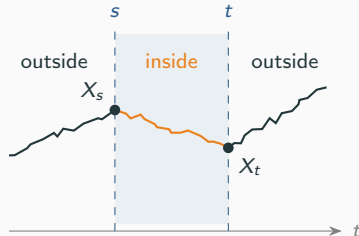
- ▶ Draw $(X_0, X_1) \sim \pi$ for **any** coupling π , then join by a Brownian bridge.
- ▶ The set of all of them is the **reciprocal class** $\mathfrak{R}(R^\varepsilon)$.
- ▶ Examples: Brownian motion itself; any Brownian bridge; the Schrödinger bridge ($\pi = \gamma_\varepsilon$).

Markov processes

- ▶ (Bernstein '32, Jamison '74) X is **reciprocal** if for all $s < t$:

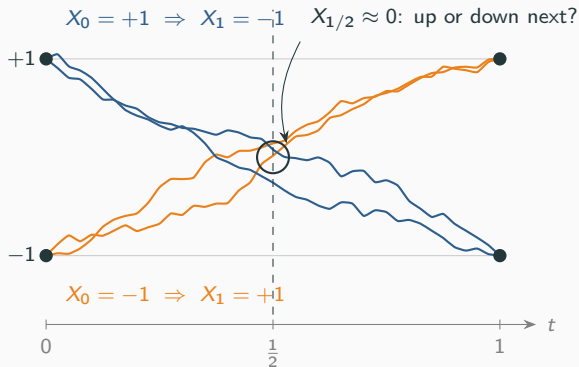
$$(X_r)_{r \in [s, t]} \perp (X_r)_{r \notin [s, t]} \mid (X_s, X_t).$$

- ▶ Markov: past $\perp$ future given the present.
Reciprocal: inside $\perp$ outside given the boundary.
- ▶ Markov $\Rightarrow$ reciprocal.



Reciprocal but not Markov

- ▶ $\mu = \nu = \frac{1}{2}\delta_{-1} + \frac{1}{2}\delta_1$. The “anti-coupling”: $X_1 = -X_0$. Join by Brownian bridge.
- ▶ Reciprocal, but **not Markov**: at time $t = \frac{1}{2}$, a path near 0 remembers where it started, and that determines where it ends.



Given $X_{1/2}$ alone the future is ambiguous; given X_0 it is determined: **not Markov**.

Theorem (Jamison '74–'75; Léonard '14)

Under mild conditions, for P with $X_0 \sim \mu$, $X_1 \sim \nu$:

P is Reciprocal and P is Markov $\iff P$ is Schrödinger bridge.

Schrödinger bridge = reciprocal $\cap$ Markov $\cap$ {correct marginals}.

Markov projections (mimicking)

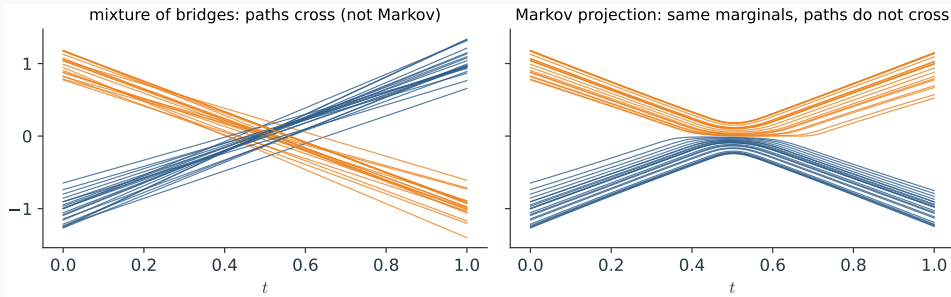
- ▶ (Gyöngy '86, Brunick & Shreve '13) $dX_t = \beta_t dt + \sqrt{\varepsilon} dW_t$, where β_t may depend on the whole path. Set

$$b_t(x) := \mathbb{E}[\beta_t \mid X_t = x], \quad dY_t = b_t(Y_t) dt + \sqrt{\varepsilon} dW_t, \quad Y_0 \stackrel{d}{=} X_0.$$

- ▶ Then $Y_t \stackrel{d}{=} X_t$ for every t : **same one-time marginals**, and Y is Markov.
- ▶ For a mixture of bridges, $\beta_t = \frac{X_1 - X_t}{1-t}$ (the bridge drift), so

$$b_t(x) = \mathbb{E}\left[\frac{X_1 - X_t}{1-t} \mid X_t = x\right].$$

Markov projection in pictures



- ▶ The anti-coupling example with slightly spread endpoints, $\varepsilon \rightarrow 0$.
- ▶ Left: the mixture of bridges; paths cross. Right: its Markov projection; the same $\text{Law}(X_t)$ for each t , but paths cannot cross, so the endpoint coupling changes.

Markov projections: variational view and ML

- ▶ Y is the **KL-projection** of X onto Markov diffusions:

$$\text{Law}(Y) = \arg \min \{ \text{KL}(\text{Law}(X) \mid Q) : Q \text{ Markov} \}.$$

- ▶ The drift is learnt by **regression**, since a conditional expectation is a least-squares predictor:

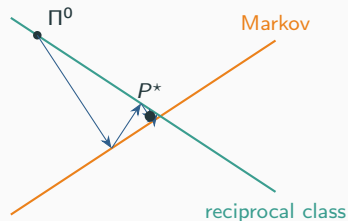
$$b = \arg \min_u \mathbb{E} \int_0^1 \left\| u_t(X_t) - \frac{X_1 - X_t}{1 - t} \right\|^2 dt.$$

- ▶ Starting from independent (X_0, X_1) and $\varepsilon = 0$: flow matching, rectified flow, stochastic interpolants in generative modelling.

Shi, De Bortoli, Campbell & Doucet (2023); Peluchetti (2023); Liu et al. (2023); Lipman et al. (2023).

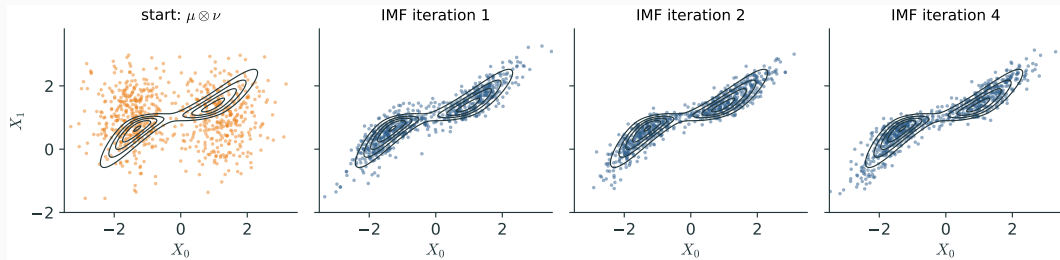
Iterated Markovian fitting (IMF)

- ▶ Start with a reciprocal process with correct endpoints, e.g. (X_0, X_1) independent. Alternate:
 1. **Markov projection**: $\Pi^{2n+1} = M(\Pi^{2n})$;
 2. **reciprocal projection**: keep the law of (X_0, X_1) , refill with Brownian bridges.
- ▶ Every iterate has $X_0 \sim \mu, X_1 \sim \nu$.
- ▶ **Theorem** (Shi et al. '23; Peluchetti '23). Under assumptions, $\text{KL}(\Pi^n \mid P^*)$ is nonincreasing and $\Pi^n \rightarrow P^*$.
- ▶ For rates see Silveri, Conforti and Durmus '25.



Same picture as Sinkhorn, different sets.

IMF in pictures



Endpoint couplings (X_0, X_1) of the IMF iterates ($\varepsilon = 0.1$, 2000 particles, drift fitted by kernel regression). Contours: γ_ε computed by Sinkhorn. Two or three iterations already get close.

Sinkhorn versus IMF

	Sinkhorn (static)	IMF (dynamic)
projects onto	$\{X \sim \mu\}, \{Y \sim \nu\}$	Markov laws, reciprocal class
always true along the way	product form	correct marginals μ, ν
true only in the limit	correct marginals	Markov $\cap$ reciprocal
needs	the cost matrix	samples + regression

- ▶ Both are alternating KL projections, and both converge to the Schrödinger bridge, static for Sinkhorn and dynamic for IMF.
- ▶ Low temperature is hard for both: Sinkhorn suffers from numerical instability. At $\varepsilon = 0$ IMF becomes rectified flow, which in $d \geq 2$ need not reach the OT coupling.

An extremely short review of statistical issues

- ▶ Estimate from samples $X_1, \dots, X_n \sim \mu$, $Y_1, \dots, Y_n \sim \nu$, with empirical laws $\hat{\mu}_n = \frac{1}{n} \sum_i \delta_{X_i}$, $\hat{\nu}_n = \frac{1}{n} \sum_j \delta_{Y_j}$.
- ▶ (Fournier & Guillin '15) $\mathbb{W}_2^2(\hat{\mu}_n, \hat{\nu}_n) \rightarrow \mathbb{W}_2^2(\mu, \nu)$ at rate $O(n^{-2/d})$ in high dimension. **Curse of dimensionality.**
- ▶ (Mena & Niles-Weed '19) For sub-Gaussian μ, ν , $\text{EOT}_\varepsilon(\hat{\mu}_n, \hat{\nu}_n)$ converges at the parametric rate $O(n^{-1/2})$ (constants depend on ε). See also Stromme '23.
- ▶ CLTs: González-Sanz, Loubes & Niles-Weed '22. Large deviations: not known.

References (1/3)

J. Altschuler, J. Weed, P. Rigollet. Near-linear time approximation algorithms for optimal transport via Sinkhorn iteration. *NeurIPS* 2017.

A. Beurling. An automorphism of product measures. *Ann. Math.* 72 (1960).

G. Brunick, S. Shreve. Mimicking an Itô process by a solution of a stochastic differential equation. *Ann. Appl. Probab.* 23 (2013).

Y. Chen, T. Georgiou, M. Pavon. On the relation between optimal transport and Schrödinger bridges: a stochastic control viewpoint. *JOTA* 169 (2016).

S. Chewi, J. Niles-Weed, P. Rigollet. *Statistical Optimal Transport*. Lecture notes, 2024.

G. Conforti, L. Tamanini. A formula for the time derivative of the entropic cost and applications. *J. Funct. Anal.* 280 (2021).

I. Csiszár. I -divergence geometry of probability distributions and minimization problems. *Ann. Probab.* 3 (1975).

M. Cuturi. Sinkhorn distances: lightspeed computation of optimal transport. *NeurIPS* 2013.

N. Deb, Y.-H. Kim, S. Pal, G. Schiebinger. Wasserstein mirror gradient flow as the limit of the Sinkhorn algorithm. arXiv:2307.16421, 2023. The Annals of Probability 54 (5) 2026.

H. Föllmer. Random fields and diffusion processes. *École d'Été de Saint-Flour XV–XVII*, LNM 1362, 1988.

References (2/3)

R. Fortet. Résolution d'un système d'équations de M. Schrödinger. *J. Math. Pures Appl.* 19 (1940).

N. Fournier, A. Guillin. On the rate of convergence in Wasserstein distance of the empirical measure. *PTRF* 162 (2015).

J. Franklin, J. Lorenz. On the scaling of multidimensional matrices. *Linear Algebra Appl.* 114/115 (1989).

I. Gentil, C. Léonard, L. Ripani. About the analogy between optimal transport and minimal entropy. *Ann. Fac. Sci. Toulouse* 26 (2017).

I. Gyöngy. Mimicking the one-dimensional marginal distributions of processes having an Itô differential. *PTRF* 71 (1986).

B. Jamison. Reciprocal processes. *Z. Wahrsch. verw. Gebiete* 30 (1974).

B. Jamison. The Markov processes of Schrödinger. *Z. Wahrsch. verw. Gebiete* 32 (1975).

C. Léonard. From the Schrödinger problem to the Monge–Kantorovich problem. *J. Funct. Anal.* 262 (2012).

C. Léonard. A survey of the Schrödinger problem and some of its connections with optimal transport. *Discrete Contin. Dyn. Syst.* 34 (2014).

Y. Lipman, R. Chen, H. Ben-Hamu, M. Nickel, M. Le. Flow matching for generative modeling. *ICLR* 2023.

X. Liu, C. Gong, Q. Liu. Flow straight and fast: learning to generate and transfer data with rectified flow. *ICLR* 2023.

References (3/3)

G. Mena, J. Niles-Weed. Statistical bounds for entropic optimal transport: sample complexity and the central limit theorem. *NeurIPS* 2019.

T. Mikami. Monge's problem with a quadratic cost by the zero-noise limit of h -path processes. *PTRF* 129 (2004).

M. Nutz. *Introduction to Entropic Optimal Transport*. Lecture notes, 2021.

S. Pal. On the difference between entropic cost and the optimal transport cost. arXiv:1905.12206, 2019. Appeared in The Annals of Probability 34 1B (2024).

S. Peluchetti. Diffusion bridge mixture transports, Schrödinger bridge problems and generative modeling. *JMLR* 24 (2023).

G. Peyré, M. Cuturi. Computational optimal transport. *Found. Trends Mach. Learn.* 11 (2019).

A.-A. Pooladian, J. Niles-Weed. Entropic estimation of optimal transport maps. arXiv:2109.12004, 2021.

L. Rüschendorf, W. Thomsen. Note on the Schrödinger equation and I -projections. *Statist. Probab. Lett.* 17 (1993).

E. Schrödinger. Über die Umkehrung der Naturgesetze. *Sitzungsber. Preuss. Akad. Wiss.* (1931).

Y. Shi, V. De Bortoli, A. Campbell, A. Doucet. Diffusion Schrödinger bridge matching. *NeurIPS* 2023.

M. Gentiloni Silveri, Giovanni Conforti, Alain Durmus. Exponential Convergence Guarantees for Iterated Markovian Fitting. Arxiv preprint 2510.20871v1 [stat.ML] (2025).

See you next time!

Lecture 3: Wasserstein space and Otto calculus